CLASSROOM CONTACT PROGRAMME
(Academic Session : 2025 - 2026)

ENTHUSIAST, LEADER & ACHIEVER COURSE
PHASE : ALL PHASE
TARGET : PRE-MEDICAL 2026

Test Type : **MAJOR** Test Pattern : **NEET (UG)**

TEST DATE : 18-05-2026

ANSWER KEY

Q.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
A.	2	4	2	4	4	3	4	1	2	4	2	1	3	1	2	1	1	2	3	1	1	3	4	3	4	1	4	1	3	1
Q.	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60
A.	1	3	1	3	3	1	1	1	4	2	3	4	2	4	1	3	1	4	3	1	2	1	2	4	2	1	1	2	3	3
Q.	61	62	63	64	65	66	67	68	69	70	71	72	73	74	75	76	77	78	79	80	81	82	83	84	85	86	87	88	89	90
A.	1	3	3	2	3	4	3	4	4	4	4	3	3	2	3	2	3	1	1	3	3	2	4	4	1	2	1	4	1	3
Q.	91	92	93	94	95	96	97	98	99	100	101	102	103	104	105	106	107	108	109	110	111	112	113	114	115	116	117	118	119	120
A.	4	2	3	2	1	3	2	1	1	2	4	2	4	3	3	3	3	3	2	2	2	4	3	3	1	1	1	3	2	2
Q.	121	122	123	124	125	126	127	128	129	130	131	132	133	134	135	136	137	138	139	140	141	142	143	144	145	146	147	148	149	150
A.	4	3	3	3	2	3	4	1	3	4	4	1	2	2	3	4	2	4	4	3	3	4	2	3	1	4	1	4	4	3
Q.	151	152	153	154	155	156	157	158	159	160	161	162	163	164	165	166	167	168	169	170	171	172	173	174	175	176	177	178	179	180
A.	3	1	3	2	1	2	4	2	3	3	1	2	2	2	1	3	3	2	4	3	3	3	3	3	3	4	3	4	2	2

HINT – SHEET

1. Ans (2)

First we determine that the total dimension of $(A + \sqrt{x})$ is $[L^{1/2}]$

Substituting the dimensions into the equation,

$$U = \frac{Bx^2}{A + \sqrt{x}}$$

$$[ML^2T^{-2}] = \frac{[B][L^2]}{[L^{1/2}]}$$

$$[B] = [M\,L^{1/2}\,T^{-2}]$$

2. Ans (4)

Here, $I \propto M^1R^2$

So,

$$\left(\frac{\Delta I}{I} \times 100\%\right) = +1\left(\frac{\Delta M}{M} \times 100\%\right) + 2\left(\frac{\Delta R}{R} \times 100\%\right)$$

$= +1\,(2\%) + 2\,(1\%)$

$= 2\% + 2\%$

$= 4\%$

3. Ans (2)

20 V.S.D. = 18 M.S.D.

$1\text{V.S.D} = \frac{18}{20}\text{M. S. D.} = 0.9\,\text{M. S. D.}$

L.C. = 1 MSD − 1 VSD

= 1 MSD − 0.9 MSD

$= 0.1\ \text{MSD} \quad \left[1\ \text{MSD} = \frac{1}{10}\text{cm}\right]$

$\boxed{\text{L. C.} = 0.01\ \text{cm}}$

4. Ans (4)

Given resistance, R = 10.932 Ω (5 significant figures) and current I = 4.25 A (3 significant figures).

Potential difference is

V = IR = 10.932 × 4.25 = 46.461 V

In multiplication, the result must have the same number of significant figures as the least precise quantity.

Here, current has 3 significant figures, so the answer must be rounded to 3 significant figures.

46.461 ⟶ 46.5 V

5. Ans (4)

$$v_{avg} = \frac{d}{\frac{2d}{5v_1} + \frac{3d}{5v_2}}$$

$$v_{avg} = \frac{5v_1v_2}{2v_2 + 3v_1}$$

6. Ans (3)

If speed, that is magnitude of velocity is constant whereas the direction of velocity changes ; we cannot say velocity constant. Therefore, the particle has non-zero acceleration.

7. Ans (4)

For first 3 seconds,

$v = u + at,$

$= 10 - (5)(3)$

$= -5 m/s$

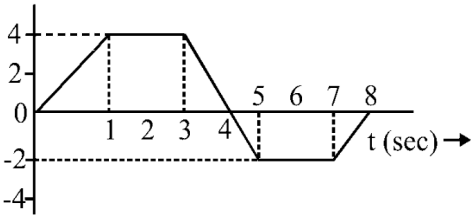
$S = (10)(3) - \frac{1}{2}(5)(3)^2$

$= 7.5 m$

Now, speed is same, $t_2 = \frac{7.5}{5} = 1.5$ sec

$T = t_1 + t_2 = 3 + 1.5 = 4.5$ sec

8. Ans (1)



Area = displacement

$= A_1 + A_2 + A_3 - A_4 - A_5 - A_6$

$= \frac{1}{2}(1 \times 4) + 2 \times 4 + \frac{1}{2} \times 1 \times 4 - \frac{1}{2} \times 1 \times 2 - 2 \times 2 - \frac{1}{2} \times 1 \times 2$

$= 2 + 8 + 2 - 1 - 4 - 1$

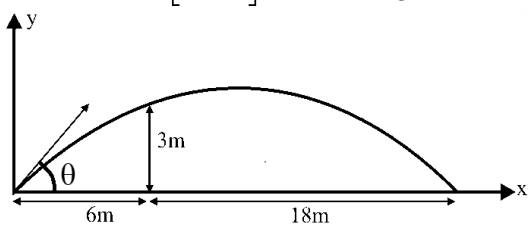
$= 12 - 6$

$= 6m$

9. Ans (2)

From equation of trajectory, $y = x\tan\theta \left[1 - \frac{x}{R}\right]$

$\Rightarrow 3 = 6\tan\theta \left[1 - \frac{1}{4}\right] \Rightarrow \tan\theta = \frac{2}{3}$



10. Ans (4)

Velocity of man, $\vec{v}_m = 10\hat{i}$ km/hr.

Velocity of rain $= -v_r\hat{j}$

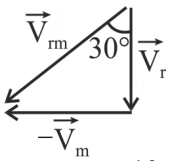
The rain appears to hit the man at an angle 30º from the vertical

The velocity of the rain with respect to the man is

$\vec{v}_{rm} = \vec{v}_r - \vec{v}_m$

The angle of v_{rm} with the vertical is 30º.

$$\tan(\theta) = \frac{|\text{ Horizontal component }|}{|\text{ Vertical component }|} = \frac{|v_m|}{|v_r|}$$



$\tan(30^o) = \frac{10}{v_r}$

$\frac{1}{\sqrt{3}} = \frac{10}{v_r}$

$v_r = 10\sqrt{3}$ km/hr

$\vec{v}_{rm} = -10\hat{i} - 10\sqrt{3}\hat{j}$ km/hr.

11. Ans (2)

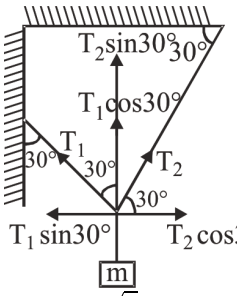
Newton's second law

$F = ma \Rightarrow 6 = (7 + 5)a ; a = \frac{1}{2}$ m/s^2 ;

for 5 kg mass,

$F' = 5 \times \frac{1}{2} = 2.5$ N

12. Ans (1)



$T_1 \sin30 = T_2 \cos30$ …(1)

$\frac{T_1}{2} = \frac{T\sqrt{3}}{2} \Rightarrow T_1 = \sqrt{3}T_2$

$T_2 \sin30 + T_1 \cos30 = mg$ …(2)

$\frac{T_1}{\sqrt{3}}\frac{1}{2} + \frac{T_1\sqrt{3}}{2} = mg$

$\frac{T_1}{2}\left[\frac{1+3}{\sqrt{3}}\right] = mg$

$T_1 = \frac{\sqrt{3}mg}{2}$

$T_2 = \frac{T_1}{\sqrt{3}} = \frac{mg}{2}$

Final Answer : option (1)

13. Ans (3)

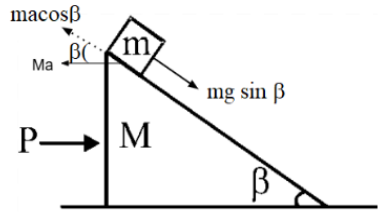
$kx = 2 \times 5$

$kx = 10$

$a = \frac{60 - kx}{4} = \frac{60 - 10}{4} = \frac{50}{4} = 12.5 m/s^2$

14. Ans (1)

$P = (M + m)\, g \tan \beta$



For block m to remain stationary relative to M, the horizontal acceleration of M must match the component of gravity along the incline for m:

$a = g \sin \beta / \cos \beta$

$a = g \tan \beta$

Force on M

P = total mass × acceleration = $(M + m) \times g \tan \beta$

15. Ans (2)

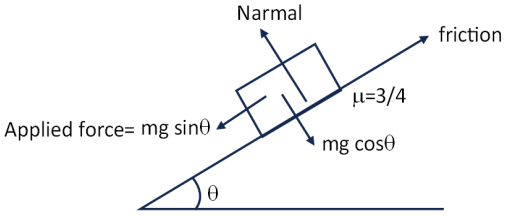
A block of mass 5 kg is placed on a 30° inclined plane with a coefficient of friction $\mu = \frac{3}{4}$. We need to determine the friction force acting on the block.

Concept-Based Explanation :-

The force of friction f is given by $f = \mu N$, where N is the normal force exerted by the surface. Normal force is calculated as $N = mg\cos\theta$.

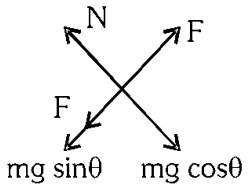
Formula used.

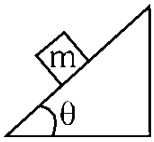
$N = mg\cos\theta$



$f = \mu N \geq F_{static}$ = Applied force = 25 N

16. Ans (1)





$F = f + mg\sin\theta = mg(\mu\cos\theta + \sin\theta)$

17. Ans (1)

Explain question :- we need to find the work done by variable force $F = -5x$ when particle moves from x = 5 m to 1 m.

Concept :- work done by a force.

Solution :-

$W = \int_1 F\, dx$

$= -\int_5 5x\, dx$

$= -5\left[\frac{x^2}{2}\right]_1^5 = 60\ J$

Final answer :- correct option (1).

18. Ans (2)

$\text{Power} = \frac{W}{dt}$

$\Delta K.E. = \int_2^4 P\, dt$

$\Delta K.E. = \int_2^4 (3t^2 - 2t + 1)\, dt$

$\Delta K.E. = \left[3 + \frac{t^3}{3} - \frac{2t^2}{2} + t\right]_2^4$

$\Delta K.E. = [64 - 16 + 4 - 8 + 4 - 2]$

$\Delta K.E. = 46\ J$

19. Ans (3)

According to work energy theorem

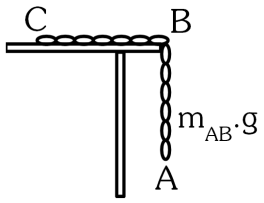
$W_{gravity} + W_{air} = \Delta KE \Rightarrow mgh + W_{air} = \frac{1}{2}mv^2 - 0$

$\Rightarrow 10[10]\,[20] + W_{air} = 500$

$\Rightarrow$ Work done by air on object

$W_{air} = -1500\ J$

20. Ans (1)

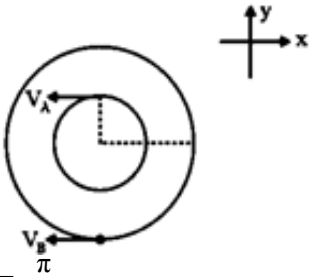


$m_{AB} = \frac{2}{5}m$

$AB = \frac{2\ell}{5}$

$W = m_{AB}g \times \frac{AB}{2} = \frac{2mgl}{25}$

21. Ans (1)

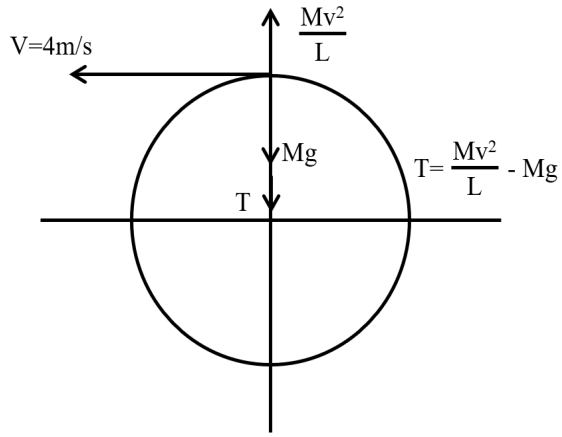


$t = \frac{\pi}{2\omega}$

$\overrightarrow{V_A} = -\omega R_1 \hat{i}$

$\overrightarrow{V_B} = -\omega R_2 \hat{i}$

22. Ans (3)



Mass m = 1 kg, velocity v = 4m/s radius r = 1 m, gravity g = 10 m/s^2.

Calculate centripetal force:

$\frac{mv^2}{r} = \frac{1 \times 4^2}{1} = 16N$

Weight mg = 10 N

from equation :

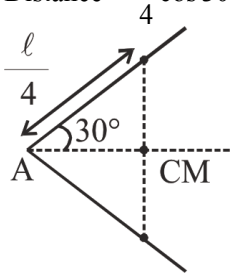
$T + 10 = 16 \Rightarrow T = 16 - 10 = 6$ N

23. Ans (4)

Direction of velocity & acceleration will change in uniform circular motion.

24. Ans (3)

Distance = $\frac{\ell}{4}\cos 30 = \frac{\ell\sqrt{3}}{8}$



25. Ans (4)

$\bar{V}_{cm} = \frac{10\left(2\hat{i} - 7\hat{j} + 3\hat{k}\right) + 2\left(-10\hat{i} + 35\hat{j} - 3\hat{k}\right)}{12} = 2\hat{k}$

26. Ans (1)

$e = \frac{|v_2 - v_1|}{|u_1 - u_2|}$

Calculation:

$m_1 = m, u_1 = V, v_1 = 0$

$m_2 = 2m, u_2 = 0$

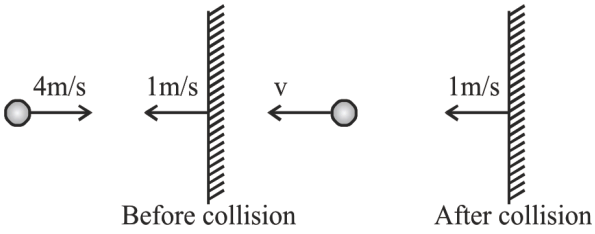
From momentum conservation:

$mV = 2m \cdot v_2 \Rightarrow v_2 = \frac{V}{2}$

Now,

$e = \frac{|v_2 - v_1|}{|u_1 - u_2|} = \frac{\left|\frac{V}{2} - 0\right|}{|V - 0|} = \frac{V/2}{V} = \frac{1}{2}$

27. Ans (4)



Let v be the velocity of ball after collision.

Collision is elastic.

$\therefore \quad e = 1$

or relative velocity of separation=relative velocity of approach

$\therefore \quad v - 1 = 4 + 1$

or v = 6 m/s (away from the wall)

28. Ans (1)

From Newton's second law,

$$F = \frac{dp}{dt}$$

If $F = 0$, then $\frac{dp}{dt} = 0$

p = constant.

Thus, if total external force acting on the system is zero, then linear momentum of the system remains conserved.

29. Ans (3)

$$\frac{2}{5}MR_1^2 = \frac{2}{3}MR_2^2 \Rightarrow \frac{R_1}{R_2} = \sqrt{\frac{5}{3}}$$

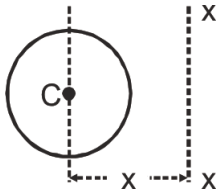
30. Ans (1)

$$\vec{\tau} = \vec{r} \times \vec{F}$$

$$\vec{\tau} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 7 & 3 & 1 \\ -3 & 1 & 5 \end{vmatrix} = \hat{i}(3\times5 - 1\times1) - \hat{j}(7\times5 - 1\times(-3)) + \hat{k}(7\times1 - 3\times(-3))$$

$$\vec{\tau} = 14\hat{i} - 38\hat{j} + 16\hat{k}$$

31. Ans (1)



$I_{xx} = I = I_c + mx^2$ (m = mass of sphere) $I = I_c$ at $x = 0$

$\therefore$ I–x graph is a parabola with minimum value of $I = I_c$ at $x = 0$.

$\therefore$ the correct graph is (a).

32. Ans (3)

Given that; $\omega = 2 + 6t$ rad/s

$$\alpha = \frac{d\omega}{dt} = 6 \text{ rad/s}^2$$

$$\tau = l\alpha = \frac{MR^2}{2}\alpha = \frac{5\times(0.2)^2\times6}{2} = 0.6 \text{ N-m}$$

Now, $\tau = FR = 0.6$

$F = 3$ N.

33. Ans (1)

Initial Angular momentum

$$L_i = I_1\omega_1 = 100\left(\frac{2\pi}{10}\right) \quad ....(1)$$

Final Angular momentum

$L_f = (I_1 + mR^2)\omega_2 = (100 + 50\times2^2)\omega_2$ ....(2)

By $L_f = L_i$

$$\Rightarrow \quad 300\omega_2 = 100\left(\frac{2\pi}{10}\right)$$

$$\Rightarrow \quad \omega_2 = \frac{2\pi}{30}$$

34. Ans (3)

$$a_r = \frac{g\sin\theta}{1 + k^2/R^2}$$

$$a = \frac{g\sin\theta}{\left(1 + \frac{K^2}{R^2}\right)} = \frac{g\sin\theta}{\left(1 + \frac{2}{5}\right)}$$

$$= \frac{5}{7}g\sin\theta$$

35. Ans (3)

Pure rolling is a combination of pure translation and pure rotation. For it to occur without slipping on a stationary ground, two conditions must be met:

1. The velocity of the center of mass must be related to the angular velocity by $V_{cm} = R\omega$.
2. The acceleration of the center of mass must be related to the angular acceleration by $a_{cm} = R\alpha$. Both statements are the fundamental conditions for pure rolling.

(3) Both (1) and (2)

36. Ans (1)

Orbital radius, $r = R + 2R = 3R$

Total mechanical energy in orbit, $E_f = -GMm/2r$

$E_f = -GMm/6R$

Initial energy on surface, $E_i = -GMm/R$

Required Energy, $\Delta E = E_f - E_i$

$\Delta E = -GMm/6R - (-GMm/R)$

$\Delta E = GMm\,(1 - 1/6)\,R = 5GMm/6R$

37. Ans (1)

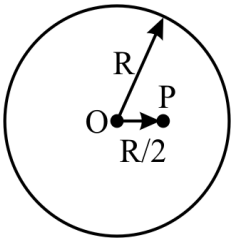
As, $v_e = \sqrt{\frac{2GM}{R}}$

$\Rightarrow \quad v_e = \sqrt{\frac{8}{3}G\pi\rho R^2}$

$\Rightarrow \quad v_e \propto R\sqrt{\rho}$

$\Rightarrow \quad v_e = v\sqrt{2}$

38. Ans (1)



Here, mass of the particle = M

Mass of the spherical shell = M

Radius of the spherical shell = R

Let O be centre of spherical shell.

Gravitational potential at point P due to particle at

O is $V_1 = -\frac{GM}{(R/2)}$

Gravitational potential at point P due to spherical

shell is $V_2 = -\frac{GM}{R}$

Hence, total gravitational potential at the point P is

$V = V_1 + V_2$

$= (-2-1)\frac{GM}{R} = -\frac{3GM}{R}$

39. Ans (4)

$g_s = \frac{4}{3}\pi G\rho R$

$g_s \propto \rho R$

$\frac{(g_s)_1}{(g_s)_2} = \frac{\rho_1 R_1}{\rho_2 R_2}$

$\frac{(g_s)_1}{(g_s)_2} = \left(\frac{1}{2}\right)\left(\frac{4}{1}\right)$

$\frac{(g_s)_1}{(g_s)_2} = \frac{2}{1}$

40. Ans (2)

$3P = P + \rho gh$

$h = \frac{2P}{\rho g}$ ..... (1)

$P^1 = P + \rho g\frac{4h}{5}$

from equation (1)

$P^1 = P + \rho g\frac{4}{5}\left(\frac{2P}{\rho g}\right)$

$P^1 = \frac{13P}{5}$

41. Ans (3)

$Mg = \rho_\omega V_{dis} g$

$V_{dis} = \frac{M}{\rho_\omega}$ (same for both A and B)

42. Ans (4)

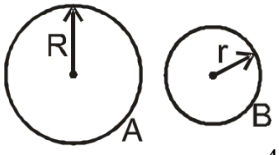
$v_t = \frac{2r^2(\rho - \sigma)g}{9\eta}$

$r^2 = \frac{9\eta v_t}{2(\rho - \sigma)g}$

$\left(\frac{r_1}{r_2}\right)^2 = \frac{\rho_2 - \sigma}{\rho_1 - \sigma} = \frac{(11-2)\times 10^3}{(8-2)\times 10^3} = \frac{9}{6} = \frac{3}{2}$

$\frac{r_1}{r_2} = \sqrt{\frac{3}{2}}$

43. Ans (2)



By equating volume : $\frac{4}{3}\pi R^3 = 8 \times \frac{4}{3}\pi r^3$

we get, r = R/2.

Now pressure difference in A = $\frac{2T}{R}$ and that in

B = $\frac{2T}{R/2}$ = 2 × pressure difference in A.

$\Rightarrow (\Delta P)_B = 2\,(\Delta P)_A$

$(\Delta P)_A = \frac{1}{2}\{(\Delta P)_B\}$

44. Ans (4)

$F = \frac{1}{2}\rho\left(V_2^2 - V_1^2\right)A$

$= \frac{1}{2} \times 1.28 \times \left((140)^2 - (110)^2\right) \times 15 = 72000\text{N}$

45. Ans (1)

$$w = \frac{1}{2}F\Delta\ell = \frac{1}{2}\frac{AY}{L}(\Delta\ell)^2$$

$$\frac{w_2}{w_1} = \left(\frac{R_2}{R_1}\right)^2 \times \left(\frac{L_1}{L_2}\right)$$

$$\frac{w_2}{2} = \left(\frac{2R}{R}\right)^2 \times \left(\frac{L}{L/2}\right)$$

$w_2 = 16J$

46. Ans (3)

$$mvr = \frac{nh}{2\pi}$$

$$\therefore \frac{2h}{\pi} = \frac{nh}{2\pi}$$

$n = y$

$\therefore$ K.E. = +0.85 eV

47. Ans (1)

$$\frac{\Delta x_A \cdot \Delta V_A}{\Delta x_B \cdot \Delta V_B} = \frac{m_B}{m_A}$$

$$\frac{\Delta x_A}{\Delta x_B} \times \frac{0.05}{0.02} = \frac{5m_B}{m_B}$$

$$\frac{\Delta x_A}{\Delta x_B} = 2$$

48. Ans (4)

Explain Question : Determine correctness of assertion–reason about electron in atom

Concept : Heisenberg uncertainty principle & probability interpretation of $|\Psi|^2$.

Solution :

Assertion (A): Path of electron cannot be determined accurately due to Heisenberg uncertainty — **Incorrect**.

Reason (R): $|\psi|^2$ gives probability distribution of electron around nucleus — **Correct**.

Final Answer : option (4).

49. Ans (3)

$$OAM = \frac{h}{2\pi}\sqrt{\ell(\ell+1)}$$

$\ell = 1$

$$\therefore OAM = \frac{h}{\sqrt{2}\pi}$$

50. Ans (1)

$$A(g) \rightleftharpoons 2B(g)$$

n_i 0.2 0

n_{eq} 0.2 −x 2x

$P_{eq}V = n_{eq}RT$

$$8 \times 1 = (0.2 + x) \times \frac{1}{12} \times 300$$

$x = 0.12$

$n_A = 0.08$ $n_B = 0.24$

$P_A = X_A \cdot P_T = 2$ bar

$P_B = X_B \cdot P_T = 6$ bar

$$K_P = \frac{(P_B)^2}{P_A} = \frac{36}{2} = 18$$

51. Ans (2)

Answer: (2)

Explanation: For a given reaction, Favourable Conditions are to be found.

Given data: $\Delta H = -21 \cdot 9$ Kcal

Concept:

- For Endothermic Reaction :- $\tau \alpha \kappa$
- For Exothermic Reaction! :- $\tau \alpha \frac{1}{k}$
- Pressure ↑ Equilibrium shifts From Higher gaseous moles to lower gases moles.
- In Presence of catalyst = Rate of reaction ↑.

Calculation:

$N_2 + 3H_2 \rightleftharpoons 2NH_3 : \Delta Hc = -21.9$

(g) (g) (g)

Final answer:

Exothermic Reaction:- T ↓ = Equilibrium Shifts in Forward direction.

Pressure ↑ Equilibrium Shifts in Forward direction

Catalyst = Equilibrium Shifts in Forward direction

Final Answer:- 2nd option.

52. Ans (1)

(i) AB type salt; $k_{sp} = s^2$

(ii) AB_2 type salt; $k_{sp} = 4\,s^3$

(iii) AB_3 type salt; $k_{sp} = 27\,s^4$

(iv) A_3B_2 type salt; $k_{sp} = 108\,s^5$

53. Ans (2)

$\rightarrow S_{BaSO_4}$ in H_2O is 10^{-5}

So, $K_{sp} = (10^{-5})^2 = 10^{-10}$

$\rightarrow$ in presence of K_2SO_4

$BaSO_4 \rightleftharpoons Ba^{2+} + SO_4^{2-}$

s' s'

$K_2SO_4 \rightarrow 2K^{-1} + SO_4^{2-}$

0.1 0.1

Total $SO_4^{2-} = 0.1 + s' \approx 0.1$

$(K_{sp})_{BaSO_4} = Ba^{2+} + SO_4^{2-}$

$10^{-10} = s' \times 0.1$

$s' = 10^{-9}$ mol/L

$s' = 2.33 \times 10^{-7}$ g/L

54. Ans (4)

$CH_3COOH + NaOH \rightarrow CH_3COONa + H_2O$

$(g_{eq.})_{WA} > (g_{eq.})_{SB} \Rightarrow$ Acidic Buffer

55. Ans (2)

NaZ

WASB salt

$pH = 7 + \frac{1}{2}pKa + \frac{1}{2}\log C$

56. Ans (1)

(P) $n = \frac{6.022 \times 10^{23}}{6.023 \times 10^{23}} = 10$

1 mole $Al_2(SO_4)_3 \cdot 3H_2O \rightarrow 15$ mole O

10 mole $\rightarrow 150$ mole O

(Q) $n = \frac{90}{180} = \frac{1}{2}$

1 mole $C_6H_{12}O_6 \rightarrow 6$ mole O

$\frac{1}{2}$ mole $\rightarrow 3$ mole O

(R) $n = \frac{112}{22.4} = 5$

1 mole $SO_3 \rightarrow 3$ mole O

5 mole $\rightarrow 15$ mole O

(S) $n = \frac{54}{108} = \frac{1}{2}$

1 mole $\rightarrow 5$ mole O

$\frac{1}{2}$ mole $\rightarrow 2.5$ mole O.

57. Ans (1)

NCERT Pg. # 15

Same no. of moles

58. Ans (2)

$^{n}CaCO_3 = \frac{50}{100} = \frac{1}{2}$ mol

1 mol $CaCO_3 \rightarrow 2$mol HCl

$\frac{1}{2}$mol $CaCO_3 \rightarrow 1$ mol HCl

$w_{HCl)_{reacted}} = 1 \times 36.5 = 36.5$ g

Since HCl is 50% pure

$\therefore w_{HCl)_{required}} = 365 \times \frac{100}{50} = 73$g

59. Ans (3)

$n_{H_2} = \frac{64}{2} = 32$ mol.

$n_{o_2} = \frac{64}{32} = 2$mol

$2H_2 + O_2 \rightarrow 2H_2O$

L.R. $\frac{32}{2}$ $\frac{2}{1}$

=16 =2

$\therefore$ L.R. $\rightarrow O_2$

ER $\rightarrow H_2$

1 mol $O_2 \rightarrow 2$ mol H_2O 1 mol$O_2 \rightarrow 2$ mol H_2

2 mol $O_2 \rightarrow 4$ mol H_2O 1 mol $O_2 \rightarrow 4$ mol O_2

$W_{H_2} = 4 \times 2 = 8$g

$\therefore W_{H_2O} = 4 \times 18 = 72$g $W_{H_2)_{left}} = 64 - 8 = 56$g

60. Ans (3)

$Cr^{3+} + BrO^- \longrightarrow CrO_4^{-2} + Br^-$

3 2

$2Cr^{3+} + 3BrO^- \rightarrow 2CrO_4^{-2} + 3Br^-$

$2Cr^{3+} + 3BrO^- + 5H_2O \rightarrow 2CrO_4^{-2} + 3Br^-$

$2Cr^{3+} + 3BrO^- + 5H_2O \rightarrow 2CrO_4^{-2} + 3Br^- + 10H^+$

$2Cr^{3+} + 3BrO^- + 5H_2O + 10OH^-$

$\rightarrow 2CrO_4^{-2} + 3Br^- + 10H^+ + 10OH^-$

$2Cr^{3+} + 3BrO^- + 10OH^- \rightarrow 2CrO_4^{-2} + 3Br^- + 5H_2O$

61. Ans (1)

KO_2 , H_2O_2

Alkali metals always show +1 O.S. in its compounds

$2(+1) + 2x = 0$

$x = -1$

H_2SO_4

$2(+1) + x + 4(-2) = 0$

$x = +6$

62. Ans (3)

Isothermal process $\Rightarrow \Delta H = 0$

63. Ans (3)

Explanation :- For a given reaction $\Delta G°$ is given and equilibrium constant is to be found

Concept :- This question is based on $\Delta G° = -2.303 RT \log K_{eq}$.

Solution :- $\Delta G° = -2.030 RT \log K_{eq}$

$-13.06 \times 10^3 = -2.303 \times 8.314 \times 298 \log K_{eq}$

$$\log K_{eq} = \frac{13.06 \times 10^3}{2.303 \times 8.314 \times 298} = 2.38$$

$= K_{eq} = 10^2 \times 10^{0.38}$

$= 2.4 \times 10^2$

64. Ans (2)

$Hg_{(\ell)} \longrightarrow H_{g(g)}$

$\Delta H = \Delta H_f(P) - \Delta H_f(R)$

$= 60.8 - 0 = 60.8 \text{ kJ mol}^{-1}$

$\Delta S = 174.4 - 77.4 = 97$

$$T = \frac{\Delta H}{\Delta S} = \frac{60.8}{97} \times 1000 = 626.8k$$

65. Ans (3)

Question Explanation:

We need to find temperature of spontaneous reaction if value of ΔH and ΔS are given.

Concept:

Gibbs Free Energy Change.

Solution:

$$T_{eq^n} = \frac{\Delta H}{\Delta S}$$

$$= \frac{-49 \times 1000}{-40.2}$$

$= 1218.9 \text{ K}$

$T = 946°C$

66. Ans (4)

Explanation : Find $(\Delta_f H)_{OH^-}$

Given data :

$H_2O(\ell) \rightarrow H^+_{aq} + H^-_{(aq)} \qquad \Delta H = 57.32 \text{ kJ}$

$H_2(g) + \frac{1}{2}O_2(g) \rightarrow H_2O(\ell)$

$(\Delta H_f)_{H_2O(\ell)} = -286.2 \text{kJ}$

$(\Delta H_f^o)(H^+, aq) = 0$

Concept : $H_2O(\ell) \rightarrow H^+(aq) + OH^-(aq)$

$\Delta H = 57.32 = [\Delta H_F(H^+, aq) + \Delta Hf(OH^- aq) - \Delta H_f(H_2O, \ell)]$

Calculation : $57.32 = [O + \Delta H_F(OH^-, aq) - (-286.20)]$

$\Delta H_F(OH^-, aq) = 57.32 - 286.2$

Final Answer : -228.88 kJ

67. Ans (3)

Explanation : Find $\Delta_C H$, for the reaction.

Given data : $C_2H_5OH(\ell) + 3O_2(g) \longrightarrow 2CO_2(g) + 3H_2O(\ell)$

$q_V = \Delta E = -1364.47 \text{ kJ mol}^{-1}$, $R = 8.314 \text{ kJ mol}^{-1}$, $T = 25°C = 298 \text{ K}$

Concept : Find Δng of a reaction and put, $\Delta H = \Delta E + \Delta ng\, RT$

Calculation :

$\Delta ng = 2 - 3 = -1.$

$$\Delta H = -1364.47 + (-1)\left(\frac{8.314}{1000}\right) \times 298$$

Final Answer : $-1366.94 \text{ kJ mol}^{-1}$.

68. Ans (4)

$\frac{1}{2}H_2 + \frac{1}{2}I_2 \rightarrow HI$

$\Delta H_r = \text{B.E. (R)} - \text{B.E (P)}$

$= \frac{1}{2}x + \frac{1}{2}y - z$

$= \frac{x}{2} + \frac{x}{2} - z$

69. Ans (4)

Question Explanation: We need to identify correct **nature (metal/non-metal/metalloid)** and **physical state at room temperature** for given atomic numbers.

Concept:

• Use periodic table trends:

• Metals ⟶ mostly solid (except some like Hg, Cs partially liquid near RT)

• Non-metals ⟶ can be gas/liquid/solid

• Metalloids ⟶ solid

• Special cases:

• Br (Z = 35) ⟶ only liquid non-metal

• Ga, Cs ⟶ low melting metals

Solution:

(a) Z = 55 ⟶ Cs

• Nature: Metal (P)

• State: Liquid (x , m.p. ≈ 28°C)

(b) Z = 31 ⟶ Ga

• Nature: Metal (P)

• State: Liquid (y)

(c) Z = 35 ⟶ Br

• Nature: Non-metal (Q)

• State: Liquid (y)

(d) Z = 52 ⟶ Te

• Nature: Metalloid (R)

• State: Solid (x)

Correct matching sequence :

(a)⟶**P, y;** (b)⟶**P, y;** (c)⟶**Q, y;** (d)⟶**R, x**

Final Answer: Option (4)

70. Ans (4)

Correct ΔHeg for 16 group

S > Se > Te > O

71. Ans (4)

Concept Based

72. Ans (3)

O > F > N > C

73. Ans (3)

Li Be B C N

1.0 1.5 2.0 2.5 3.0 ⟶

Z_{eff} ↑ EN ↑

74. Ans (2)

$Is^2, 2s^2, 2p^6, 3s^2, 3p^6, 4s^1$ ⟶ Alkali metal.

and it form Basic Oxide.

75. Ans (3)

Z = 118 = oganesson Og

76. Ans (2)

EA : Cl > O > C > N

ΔHeg_1 of N is positive

77. Ans (3)

Explaion Question: Question is asking about incorrect option regarding Lewis dot structure

Concept: Lewis dot structure, Formal charge, Resonance

Solution: The lowest energy structure is one with the lowest formal charge on the atoms.

Final Answer: 3

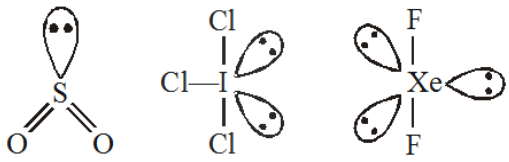
78. Ans (1)

New NCERT XI, Pg. # 119 Part-I

79. Ans (1)

	Hybridization (संकरण)
IF_5	sp^3d^2
I_3^-	sp^3d
I_3^+	sp^3

80. Ans (3)



81. Ans (3)

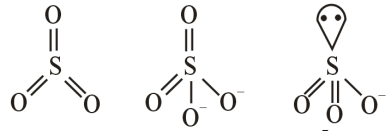
$Cl_2 > Br_2 > F_2 > I_2$ (Bond energy)

82. Ans (2)

(−1) (+1) (−1)

N = N = N

83. Ans (4)



Bond order = 2 1.5 $\frac{5}{3}=1.6$

Bond order $\propto$ bond strength

84. Ans (4)

Explain Question: Find the mismatched molecule with its geometry and polarity.

Concept: Molecular geometry & dipole moment.

Solution:

- PCl_3F_2 ⟶ trigonal bipyramidal, nonplanar, nonpolar
- ClF_3 ⟶ T-shaped, planar, polar
- XeF_2 ⟶ linear, planar, nonpolar
- XeF_6 ⟶ distorted octahedral, **nonplanar and polar**

Final Answer: Option (4)

85. Ans (1)

Concept

86. Ans (2)

Acidic strength HF < HCl < HBr < HI

(K_a)

87. Ans (1)

Explain Question: Determine correctness of statements about boiling points of Group-16 hydrides.

Concept: This question is based on Hydrogen bonding and molecular mass

Solution: Boiling point order: $H_2O > H_2Te > H_2Se > H_2S$ — correct (Statement I true).

Reason: H_2O forms strong H-bonds, giving higher b.p. than expected from molecular mass — also true.

Final Answer: Option (1)

88. Ans (4)

Explain Question : Match molecular species with correct molecular orbital theory-based properties.

Concept : This question is based on MOT

Solution :

(A) N_2 : (14 electrons) - $\sigma1s^2, \sigma^*1s^2, \sigma2s^2, \sigma^*2s^2, \pi2px^2 = \pi2py^2, \sigma2pz^2$ B.O. = 3

N_2^+ : (13 electrons) - $\sigma1s^2, \sigma^*1s^2, \sigma2s^2, \sigma^*2s^2, \pi2px^2 = \pi2py^2, \sigma2pz^1$ B.O. = 2.5

(B) B_2 : (10 electrons) - $\sigma1s^2, \sigma^*1s^2, \sigma2s^2, \sigma^*2s^2, \pi2px^1 = \pi2py^1$ B.O. = 1

B_2^+ : (9 electrons) - $\sigma1s^2, \sigma^*1s^2, \sigma2s^2, \sigma^*2s^2, \pi2px^1$ B.O. = 0.5

(C) O_2 (16 electrons) - $\sigma1s^2, \sigma^*1s^2, \sigma2s^2, \sigma^*2s^2, \sigma2pz^2, \pi2px^2 = \pi2py^2, \pi^*2px^1 = \pi^*2py^1$

B.O. = 2

O_2^{2-} (18 electrons) - $\sigma1s^2, \sigma^*1s^2, \sigma2s^2, \sigma^*2s^2, \sigma2pz^2, \pi2px^2 = \pi2py^2, \pi^*2px^2 = \pi^*2py^2$

B.O. = 1

(D) H_2 (2 electrons) - $\sigma1s^2$ B.O. = 1

H_2^+ (1 electrons) - $\sigma1s^1$ B.O. = 0.5

Final Answer : option (4)

89. Ans (1)

Explain the Question :- Identify the correct statement about polarisation caused by a cation.

Concept :-This question is based on Fajan's rule (polarising power)

Solution :- Polarising power increases with higher positive charge and smaller ionic radius. Hence, a cation with high charge causes maximum polarisation.

Final answer :- (1)

90. Ans (3)

SO_2, SO_2 Both have $p\pi-p\pi$ and $p\pi-d\pi$

91. Ans (4)

NCERT-XI, Pg. # 8

92. Ans (2)

NCERT XI Pg. # 4

93. Ans (3)

NCERT-XI, Pg. # 13

94. Ans (2)

NCERT-XI, Pg. # 20, 21

95. Ans (1)

NCERT-XI, Pg. # 17

96. Ans (3)

NCERT-XI, Pg. # 13, 15, 18

97. Ans (2)

NCERT-XI, Pg. # 13

98. Ans (1)

NCERT-XI, Pg. # 18

99. Ans (1)

NCERT-XI, Pg. # 15

100. Ans (2)

NCERT-XI, Pg. # 17, 18

101. Ans (4)

NCERT-XI, Pg. # 14, 15

102. Ans (2)

NCERT-XI, Pg. # 15, 16, 19, 24

103. Ans (4)

NCERT-XI, Pg. # 30

104. Ans (3)

NCERT-XI, Pg. # 31

105. Ans (3)

NCERT-XI, Pg. # 24

106. Ans (3)

NCERT-XI, Pg. # 29

107. Ans (3)

NCERT-XI, Pg. # 26, 28, 30, 32

108. Ans (3)

NCERT-XI, Pg. # 26, 29, 32, 34

109. Ans (2)

NCERT-XI, Pg. # 25

110. Ans (2)

NCERT-XI, Pg. # 127

111. Ans (2)

NCERT-XI, Pg. # 121

112. Ans (4)

NCERT-XI, Pg. # 123

113. Ans (3)

NCERT-XI, Pg. # 127

114. Ans (3)

NCERT-XI, Pg. # 122

115. Ans (1)

NCERT-XI, Pg. # 123

116. Ans (1)

NCERT-XI, Pg. # 136, 137

117. Ans (1)

NCERT-XI, Pg. # 126, 127

118. Ans (3)

NCERT-XI, Pg. # 121, 122

119. Ans (2)

NCERT-XI, Pg. # 110

120. Ans (2)

NCERT-XI, Pg. # 106

121. Ans (4)

NCERT-XI, Pg. # 177

122. Ans (3)

NCERT-XI, Pg. # 118

123. Ans (3)

NCERT-XI, Pg. # 115

124. Ans (3)

NCERT-XI, Pg. # 109-110

125. Ans (2)

NCERT-XI, Pg # 108

126. Ans (3)

NCERT-XI, Pg. # 88

127. Ans (4)

NCERT-XI, Pg. # 97, 98

128. Ans (1)

NCERT-XI, Pg. # 100

129. Ans (3)

NCERT-XI, Pg. # 93, 94

130. Ans (4)

NCERT-XI, Pg. # 94

131. Ans (4)

NCERT-XI, Pg. # 96, 97

132. Ans (1)

NCERT-XI, Pg. # 90

133. Ans (2)

NCERT-XI, Pg. # 100

134. Ans (2)

NCERT-XI, Pg. # 88

135. Ans (3)

NCERT-XI, Pg. # 29

136. Ans (4)

NCERT, Pg. # 41

137. Ans (2)

NCERT Pg. # 47

138. Ans (4)

The correct answer is Option 4: lchthyophis

- lchthyophis belongs to the order Gymnophiona, which are commonly known as caecilians.
- Caecilians are limbless amphibians, meaning they lack legs.

139. Ans (4)

NCERT, Pg. # 43

140. Ans (3)

NCERT, Pg. # 44

141. Ans (3)

NCERT_ Pg. No. 43

142. Ans (4)

NCERT Pg. # 38-39

143. Ans (2)

Question Explanation : Identify the figures A, B, C, and D correctly.

Concept : Cnidaria and Ctenophora identification

Solution :

- A : Aurelia (jellyfish)
- B : Adamsia (sea anemone)
- C : Cnidoblast (stinging cell)
- D : Pleurobrachia (comb jelly)

Final Answer : 2

144. Ans (3)

NCERT Pg.#48-51

145. Ans (1)

NCERT, Pg # 47, 48

146. Ans (4)

NCERT, Pg. # 48

147. Ans (1)

NCERT, Pg. # 49

148. Ans (4)

Question Explanation: Identifying exclusively marine phyla.

Concept: Animal habitats.

Solution: Ctenophora and Echinodermata are phyla with exclusively marine members. Mollusca, Annelida, and Chordata also include freshwater and terrestrial species.

Final Answer: 4

149. Ans (4)

NCERT Pg. # 45

150. Ans (3)

Module No. 4, Pg. # 94

151. Ans (3)

Module-4, Pg. # 146

152. Ans (1)

Module No. 4, Pg. # 149

153. Ans (3)

NCERT Pg # 81

154. Ans (2)

NCERT, Pg. # 82,83

155. Ans (1)

NCERT-XI, Pg. # 83

156. Ans (2)

10 pairs of cranial nerves emerge, each serving specific sensory and motor functions. These nerves are responsible for various activities like vision, hearing, taste, smell, and muscle control.ans-2

157. Ans (4)

NCERT, Pg. # 196

158. Ans (2)

NCERT Pg.#202

159. Ans (3)

NCERT (XII) Page No. 196

160. Ans (3)

NCERT Pg.#195.

161. Ans (1)

NCERT Pg. # 82

162. Ans (2)

NCERT Pg. # 200

163. Ans (2)

NEW NCERT Biology XI Page no. 203

164. Ans (2)

NEW NCERT Biology XI Page no. 200

165. Ans (1)

NCERT Pg. # 187

166. Ans (3)

NCERT Pg. # 187

167. Ans (3)

NCERT Pg. # 184

168. Ans (2)

NCERT Pg. # 183

169. Ans (4)

NCERT Pg. # 189

170. Ans (3)

NCERT Pg. # 190

171. Ans (3)

NCERT XIIth, Pg.No.275 Q.17.6

172. Ans (3)

NCERT Pg. # 185

173. **Ans (3)**

XIth NCERT Page No. – 292, 293

174. **Ans (3)**

NCERT-XI, Pg. # 289, Para (19.8)

175. **Ans (3)**

NCERT Pg. # 298

176. **Ans (4)**

NCERT Pg. # 205

177. **Ans (3)**

NCERT Pg. # 207

178. **Ans (4)**

NCERT XIIth, Pg.No.291 Intro.

179. **Ans (2)**

NCERT Pg. # 212

180. **Ans (2)**

NCERT XIIth, Pg.No.293 Q.19.2